Group II, Row IV: final result ($\delta^{mm} = d_\perp$, exact limits)

$$
\begin{aligned}
\frac{d^3N_{\rm IV}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,d_\perp\left[\log\left(\frac{\vee}{\Lambda}\right)-\log\left(\frac{\vee}{k^+}\right)+\log\left(1-\frac{k^+}{\vee}\right)+\log\left(1+\frac{\Lambda}{k^+}\right)\right]\\
&\quad\times\int_{x,x',y,y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}\,e^{2\gamma_E}}{4}\right]\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,2\left[\log\left(\frac{\vee}{\Lambda}\right)-\log\left(\frac{\vee}{k^+}\right)+\log\left(1-\frac{k^+}{\vee}\right)+\log\left(1+\frac{\Lambda}{k^+}\right)\right]\\
&\quad\times\int_{x,x',y,y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,d_\perp\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\int_{x,x',y,y'}\frac{1}{1-\xi}\left[\frac{1}{\xi^2}\,e^{-ik\cdot(y'-y)/\xi}-e^{-ik\cdot(y'-y)}\right]\\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}\,e^{2\gamma_E}}{4}\right]\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,d_\perp\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi^2}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\,\xi(1-\xi)\\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}\,e^{2\gamma_E}}{4}\right]\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,d_\perp\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi^2}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\,\frac{1-\xi}{\xi}\\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}\,e^{2\gamma_E}}{4}\right]\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,d_\perp\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi^2}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2_{\overline{MS}}\,e^{2\gamma_E}}{4}\right]\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,2\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\int_{x,x',y,y'}\frac{1}{1-\xi}\left[\frac{1}{\xi^2}\,e^{-ik\cdot(y'-y)/\xi}-e^{-ik\cdot(y'-y)}\right]\\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,2\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}\frac{d\xi}{\xi^2}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\\
&\quad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\\
&\quad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)
\end{aligned}
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\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{g^4 N_c}{16\pi^4}\,\frac{1}{k^+}\,2\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ +\Lambda}}\frac{d\xi}{\xi^2}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\,\frac{1-\xi}{\xi}\\
&\qquad\times\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\\
&\qquad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\,\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{1}{(2\pi)^3}\,\frac{g^4 N_c}{16\pi^4}\,\frac{1}{k^+}\,2\,d_\perp\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ +\Lambda}}\frac{d\xi}{\xi^2}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\,\xi(1-\xi)\\
&\qquad\times\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\\
&\qquad\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\,\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^{e}(x)
\end{aligned}
\tag{1}
$$